

Programming for Deep Learning (CO3027)

Final Project Report

Topic	Robust CAPTCHA Recognition under Corrupted and Restored Input Conditions
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1. Brief Introduction, Motivation, and Contributions

This project investigates robust CAPTCHA recognition when image inputs are clean, corrupted, or restored. CAPTCHA images are commonly used for human verification, but visual noise and distortion can make recognition difficult for deep learning models. The motivation of this project is to test whether image restoration can recover useful visual structure and improve downstream CAPTCHA recognition.

The main contribution is an integrated three-module pipeline: Module A prepares corrupted and restored CAPTCHA inputs, Module B recognizes CAPTCHA strings using sequence models, and Module C provides a position-wise multi-head classifier for fixed-length CAPTCHA recognition. Together, these modules compare restoration quality, recognition accuracy, and robustness under different input conditions.

2. Project Main Content

The system compares the same CAPTCHA batch under three input states: clean/original, dirty, and restored. The clean/original input is used as the uncorrupted reference condition, dirty input is the corrupted stress-test condition, and restored input is used to check whether restoration recovers recognition performance.

Module	Main Role	Methods / Architecture	Outputs
Module A	Image restoration and data preparation	DnCNN-based restoration models with supervised, unsupervised, semi-supervised, and self-supervised settings	Clean, dirty, and restored image conditions; restoration model summaries and denoised previews
Module B	Sequence-based CAPTCHA recognition	CRNN and Transformer recognizers with CTC-style sequence learning	Sequence accuracy, character accuracy, edit distance, and recognition summaries
Module C	Position-wise fixed-length classification	Shared encoder with five classification heads; pure CNN, ResNet, and STN variants	Per-position accuracy, sequence accuracy, edit distance, failure cases, and downstream chain comparisons

3. AI Architectures and Methodologies

- Restoration pipeline: corrupted CAPTCHA images are mapped toward normal or clean targets using DnCNN-style restoration models.
- Sequence recognition: CRNN and Transformer models process the whole CAPTCHA image as a sequence without manual character segmentation.
- Position-wise recognition: Module C assumes a fixed five-character CAPTCHA string and predicts each character position using a separate classification head.
- Learning strategies: supervised learning, contrastive learning, pseudo-label semi-supervised learning, and consistency-based self-supervised learning were compared.
- Evaluation: the project uses restoration metrics, sequence accuracy, character accuracy, edit distance, failure cases, and downstream pipeline comparisons.

4. Performance Results

The submitted results folder includes summary CSV files, plots, denoised preview figures, and downstream evaluation outputs. The key observation is that restoration can move corrupted CAPTCHA images back toward a recognizable range, while sequence-based recognition and position-wise classification provide complementary views of recognition performance.

Result Area	Included Evidence
Module A	Model summary CSV and denoised preview images for M1-M8 restoration variants.
Module B	Recognition summaries, training curves, and chain-ranking charts.
Module C	Chain ranking charts, downstream evaluation summaries, failure cases, and report assets.
Large Artifacts	Oversized checkpoints and predictions are documented in the dataset/tool notes and artifact CSV files.

5. Work Assignment

Member	Main Responsibility	Concrete Work
吳睿嫻	Module A	Module A experiments, and presentation visual design.
葉俊廷	Module B	Module B recognition experiments, result analysis, and report.
盛正璿	Module C	Module C experiments, proposal planning, idea integration, file organization, task coordination, report writing and project integration.

6. Existing AI Tools Utilization

AI tools were used as support tools for organizing the submission folder, improving document and presentation formatting, checking required materials, and translating presentation text into English. The AI tools were not used as a replacement for the model training pipeline; the included experimental outputs and code artifacts come from the project implementation and generated results.

Use Case	How It Was Used
Code organization	Helped organize project files into module-level folders and identify missing submission materials.
Document support	Helped format the final report and ensure it follows the required sections.
Presentation support	Helped convert Chinese presentation text into English while preserving project meaning.
Submission checking	Helped verify that the zip folder contains the required presentation, code, results, and notes.

7. Submitted Files

- Presentation PPT: included in the presentation folder.
- Finished code: included in the code folder.
- Performance results: included in the results folder.
- Trained models, Large files and dataset: available from the GitHub repository <https://github.com/firedragon0314/captcha-recognition-robustness>.